137. **Test scores** A student needs a total of 400 points on five tests to pass a course. The student scored 82, 91, 75, 88, and 70. Did the student pass the course?

138. **Elevators** The maximum weight capacity of an elevator is 1150 pounds. Six men are in the elevator. Their weights are 210, 145, 183, 230, 159, and 164 pounds. Is the total weight below the elevator's maximum capacity?

Writing Exercises

139. How confident do you feel about your knowledge of the addition facts? If you are not fully confident, what will you do to improve your skills?

140. How have you used models to help you learn the addition facts?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use addition notation.			
model addition of whole numbers.			
add whole numbers without models.			
translate word phrases to math notation.			
add whole numbers in applications.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

1.3 Subtract Whole Numbers

Learning Objectives

By the end of this section, you will be able to:

- Use subtraction notation
- Model subtraction of whole numbers
- Subtract whole numbers
- Translate word phrases to math notation
- Subtract whole numbers in applications

☑ **BE PREPARED** 1.3 Before you get started, take this readiness quiz.

Model $3 + 4$ using base-ten blocks.
If you missed this problem, review Example 1.12.

☑ **BE PREPARED** 1.4 Add: $324 + 586$.
If you missed this problem, review Example 1.20.

Use Subtraction Notation

Suppose there are seven bananas in a bowl. Elana uses three of them to make a smoothie. How many bananas are left in the bowl? To answer the question, we subtract three from seven. When we subtract, we take one number away from another to find the **difference**. The notation we use to subtract 3 from 7 is

$$7 - 3$$

We read $7 - 3$ as *seven minus three* and the result is *the difference of seven and three*.

Subtraction Notation

To describe subtraction, we can use symbols and words.

Operation	Notation	Expression	Read as	Result
Subtraction	$-$	$7-3$	seven minus three	the difference of 7 and 3

EXAMPLE 1.26

Translate from math notation to words: (a) $8-1$ (b) $26-14$.

Solution

(a) We read this as *eight minus one*. The result is *the difference of eight and one*.

(b) We read this as *twenty-six minus fourteen*. The result is *the difference of twenty-six and fourteen*.

TRY IT 1.51 Translate from math notation to words:

(a) $12-4$ (b) $29-11$

TRY IT 1.52 Translate from math notation to words:

(a) $11-2$ (b) $29-12$

Model Subtraction of Whole Numbers

A model can help us visualize the process of subtraction much as it did with addition. Again, we will use base-10 blocks. Remember a block represents 1 and a rod represents 10. Let's start by modeling the subtraction expression we just considered, $7-3$.

We start by modeling the first number, 7.	 7
Now take away the second number, 3. We'll circle 3 blocks to show that we are taking them away.	
Count the number of blocks remaining.	
There are 4 ones blocks left.	We have shown that $7-3=4$.

EXAMPLE 1.27

Model the subtraction: $8-2$.

Solution

$8 - 2$ means the difference of 8 and 2.

Model the first, 8.	 8
Take away the second number, 2.	
Count the number of blocks remaining.	
There are 6 ones blocks left.	We have shown that $8 - 2 = 6$.

TRY IT 1.53 Model: $9 - 6$.

TRY IT 1.54 Model: $6 - 1$.

EXAMPLE 1.28

Model the subtraction: $13 - 8$.

Solution

Model the first number, 13. We use 1 ten and 3 ones.	
Take away the second number, 8. However, there are not 8 ones, so we will exchange the 1 ten for 10 ones.	 10 3
Now we can take away 8 ones.	
Count the blocks remaining.	
There are five ones left.	We have shown that $13 - 8 = 5$.

As we did with addition, we can describe the models as ones blocks and tens rods, or we can simply say ones and tens.

TRY IT 1.55 Model the subtraction: $12 - 7$.

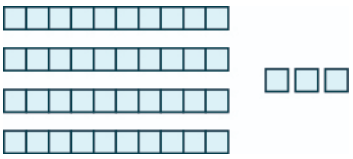
TRY IT 1.56 Model the subtraction: $14 - 8$.

EXAMPLE 1.29

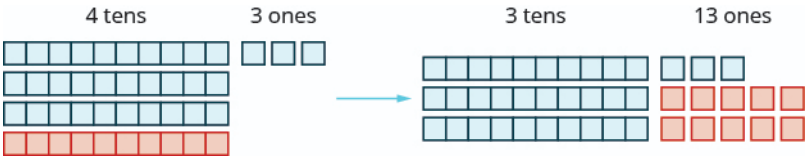
Model the subtraction: $43 - 26$.

Solution

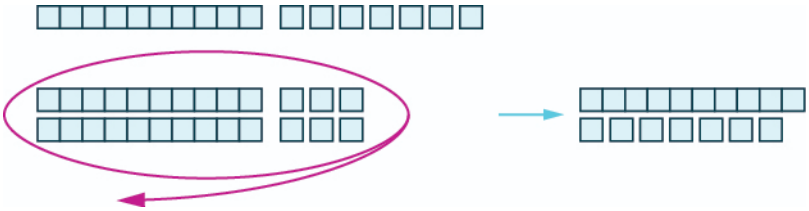
Because $43 - 26$ means 43 take away 26, we begin by modeling the 43.



Now, we need to take away 26, which is 2 tens and 6 ones. We cannot take away 6 ones from 3 ones. So, we exchange 1 ten for 10 ones.



Now we can take away 2 tens and 6 ones.



Count the number of blocks remaining. There is 1 ten and 7 ones, which is 17.

$43 - 26 = 17$

TRY IT 1.57 Model the subtraction: $42 - 27$.

TRY IT 1.58 Model the subtraction: $45 - 29$.

Subtract Whole Numbers

Addition and subtraction are inverse operations. Addition undoes subtraction, and subtraction undoes addition.

We know $7 - 3 = 4$ because $4 + 3 = 7$. Knowing all the addition number facts will help with subtraction. Then we can check subtraction by adding. In the examples above, our subtractions can be checked by addition.

$$\begin{array}{lll} 7 - 3 = 4 & \text{because} & 4 + 3 = 7 \\ 13 - 8 = 5 & \text{because} & 5 + 8 = 13 \\ 43 - 26 = 17 & \text{because} & 17 + 26 = 43 \end{array}$$

EXAMPLE 1.30

Subtract and then check by adding:

(a) $9 - 7$ (b) $8 - 3$.

Solution

(a)

	$9 - 7$

Subtract 7 from 9.	2
Check with addition. $2+7=9$✓	

ⓑ

	$8-3$
Subtract 3 from 8.	5
Check with addition. $5+3=8$✓	

TRY IT 1.59 Subtract and then check by adding:

$7-0$

TRY IT 1.60 Subtract and then check by adding:

$6-2$

To subtract numbers with more than one digit, it is usually easier to write the numbers vertically in columns just as we did for addition. Align the digits by place value, and then subtract each column starting with the ones and then working to the left.

EXAMPLE 1.31

Subtract and then check by adding: $89-61$.

Solution

Write the numbers so the ones and tens digits line up vertically.	$\begin{array}{r} 89 \\ -61 \end{array}$
Subtract the digits in each place value. Subtract the ones: $9-1=8$ Subtract the tens: $8-6=2$	$\begin{array}{r} 89 \\ -61 \\ \hline 28 \end{array}$
Check using addition. $\begin{array}{r} 28 \\ +61 \\ \hline 89 \end{array}$	

Our answer is correct.

TRY IT 1.61 Subtract and then check by adding: $86-54$.

TRY IT 1.62 Subtract and then check by adding: $99 - 74$.

When we modeled subtracting 26 from 43, we exchanged 1 ten for 10 ones. When we do this without the model, we say we borrow 1 from the tens place and add 10 to the ones place.



HOW TO

Find the difference of whole numbers.

Step 1. Write the numbers so each place value lines up vertically.
Step 2. Subtract the digits in each place value. Work from right to left starting with the ones place. If the digit on top is less than the digit below, borrow as needed.
Step 3. Continue subtracting each place value from right to left, borrowing if needed.
Step 4. Check by adding.

EXAMPLE 1.32

Subtract: $43 - 26$.

Solution

Write the numbers so each place value lines up vertically.	$\begin{array}{r} 43 \\ -\,26 \\ \hline \end{array}$
Subtract the ones. We cannot subtract 6 from 3, so we borrow 1 ten. This makes 3 tens and 13 ones. We write these numbers above each place and cross out the original digits.	$\begin{array}{r} {\scriptstyle 3\ 13} \\ \cancel{43} \\ -\,26 \\ \hline \end{array}$
Now we can subtract the ones. $13 - 6 = 7$. We write the 7 in the ones place in the difference.	$\begin{array}{r} {\scriptstyle 3\ 13} \\ \cancel{43} \\ -\,26 \\ \hline 7 \end{array}$
Now we subtract the tens. $3 - 2 = 1$. We write the 1 in the tens place in the difference.	$\begin{array}{r} {\scriptstyle 3\ 13} \\ \cancel{43} \\ -\,26 \\ \hline 17 \end{array}$
Check by adding. $\begin{array}{r} 17 \\ +26 \\ \hline 43 \checkmark \end{array}$ Our answer is correct.	

TRY IT 1.63 Subtract and then check by adding: $93 - 58$.

TRY IT 1.64 Subtract and then check by adding: $81 - 39$.

EXAMPLE 1.33

Subtract and then check by adding: $207 - 64$.

Solution

Write the numbers so each place value lines up vertically.	$\begin{array}{r} 207 \\ -\ 64 \\ \hline \end{array}$
Subtract the ones. $7 - 4 = 3$. Write the 3 in the ones place in the difference.	$\begin{array}{r} 207 \\ -\ 64 \\ \hline 3 \end{array}$
Subtract the tens. We cannot subtract 6 from 0 so we borrow 1 hundred and add 10 tens to the 0 tens we had. This makes a total of 10 tens. We write 10 above the tens place and cross out the 0. Then we cross out the 2 in the hundreds place and write 1 above it.	$\begin{array}{r} {\scriptstyle 1\ 10} \\ 2\not{0}7 \\ -\ 64 \\ \hline 3 \end{array}$
Now we subtract the tens. $10 - 6 = 4$. We write the 4 in the tens place in the difference.	$\begin{array}{r} {\scriptstyle 1\ 10} \\ 2\not{0}7 \\ -\ 64 \\ \hline 43 \end{array}$
Finally, subtract the hundreds. There is no digit in the hundreds place in the bottom number so we can imagine a 0 in that place. Since $1 - 0 = 1$, we write 1 in the hundreds place in the difference.	$\begin{array}{r} {\scriptstyle 1\ 10} \\ 2\not{0}7 \\ -\ 64 \\ \hline 143 \end{array}$
Check by adding. $\begin{array}{r} {\scriptstyle 1} \\ 143 \\ +\ 64 \\ \hline 207\checkmark \end{array}$ Our answer is correct.	

TRY IT 1.65 Subtract and then check by adding: $439 - 52$.

TRY IT 1.66 Subtract and then check by adding: $318 - 75$.

EXAMPLE 1.34

Subtract and then check by adding: $910 - 586$.

Solution

Write the numbers so each place value lines up vertically.	$\begin{array}{r} 910 \\ -\ 586 \\ \hline \end{array}$
Subtract the ones. We cannot subtract 6 from 0, so we borrow 1 ten and add 10 ones to the 0 ones we had. This makes 10 ones. We write a 0 above the tens place and cross out the 1. We write the 10 above the ones place and cross out the 0. Now we can subtract the ones. $10 - 6 = 4$.	$\begin{array}{r} {\scriptstyle 0\ 10} \\ 9\not{1}\not{0} \\ -\ 586 \\ \hline \end{array}$

Write the 4 in the ones place of the difference.	$\begin{array}{r} {}^{0}\not{9}\,{}^{10}\not{1}\not{0} \\ -\ 5\,8\,6 \\ \hline 4 \end{array}$
Subtract the tens. We cannot subtract 8 from 0, so we borrow 1 hundred and add 10 tens to the 0 tens we had, which gives us 10 tens. Write 8 above the hundreds place and cross out the 9. Write 10 above the tens place.	$\begin{array}{r} {}^{8}\not{9}\,{}^{10}\not{1}\not{0} \\ -\ 5\,8\,6 \\ \hline 4 \end{array}$
Now we can subtract the tens. $10 - 8 = 2$.	$\begin{array}{r} {}^{8}\not{9}\,{}^{10}\not{1}\not{0} \\ -\ 5\,8\,6 \\ \hline 2\,4 \end{array}$
Subtract the hundreds place. $8 - 5 = 3$ Write the 3 in the hundreds place in the difference.	$\begin{array}{r} {}^{8}\not{9}\,{}^{10}\not{1}\not{0} \\ -\ 5\,8\,6 \\ \hline 3\,2\,4 \end{array}$
Check by adding. $\begin{array}{r} {}^{1}3\,{}^{1}2\,4 \\ +\ 5\,8\,6 \\ \hline 9\,1\,0 \end{array}$ ✓	
Our answer is correct.	

TRY IT 1.67 Subtract and then check by adding: $832 - 376$.

TRY IT 1.68 Subtract and then check by adding: $847 - 578$.

EXAMPLE 1.35

Subtract and then check by adding: $2{,}162 - 479$.

Solution

Write the numbers so each place value lines up vertically.	$\begin{array}{r} 2,1\,6\,2 \\ -\ 4\,7\,9 \end{array}$
Subtract the ones. Since we cannot subtract 9 from 2, borrow 1 ten and add 10 ones to the 2 ones to make 12 ones. Write 5 above the tens place and cross out the 6. Write 12 above the ones place and cross out the 2.	$\begin{array}{r} 2,1\,{}^{5}\not{6}\,{}^{12}\not{2} \\ -\ 4\,7\,9 \end{array}$
Now we can subtract the ones.	$12 - 9 = 3$
Write 3 in the ones place in the difference.	$\begin{array}{r} 2,1\,{}^{5}\not{6}\,{}^{12}\not{2} \\ -\ 4\,7\,9 \\ \hline 3 \end{array}$

Subtract the tens. Since we cannot subtract 7 from 5, borrow 1 hundred and add 10 tens to the 5 tens to make 15 tens. Write 0 above the hundreds place and cross out the 1. Write 15 above the tens place.	$\begin{array}{r} {\scriptstyle 0\ 15\ 12} \\ 2,\not{1}\not{6}\not{2} \\ -\ 4\,7\,9 \\ \hline 3 \end{array}$
Now we can subtract the tens.	$15-7=8$
Write 8 in the tens place in the difference.	$\begin{array}{r} {\scriptstyle 0\ 15\ 12} \\ 2,\not{1}\not{6}\not{2} \\ -\ 4\,7\,9 \\ \hline 8\,3 \end{array}$
Now we can subtract the hundreds.	$\begin{array}{r} {\scriptstyle 1\ 10\ 15\ 12} \\ \not{2},\not{1}\not{6}\not{2} \\ -\ 4\,7\,9 \\ \hline 8\,3 \end{array}$
Write 6 in the hundreds place in the difference.	$\begin{array}{r} {\scriptstyle 1\ 10\ 15\ 12} \\ \not{2},\not{1}\not{6}\not{2} \\ -\ 4\,7\,9 \\ \hline 6\,8\,3 \end{array}$
Subtract the thousands. There is no digit in the thousands place of the bottom number, so we imagine a 0. $1-0=1$. Write 1 in the thousands place of the difference.	$\begin{array}{r} {\scriptstyle 1\ 10\ \not{1}\not{5}\ \not{1}\not{2}} \\ \not{2},\not{1}\not{6}\not{2} \\ -\ 4\,7\,9 \\ \hline 1,6\,8\,3 \end{array}$
Check by adding. $\begin{array}{r} {\scriptstyle 1\ \ 1\ 1} \\ 1,683 \\ +\ \ 479 \\ \hline 2,\ 162\checkmark \end{array}$	

Our answer is correct.

TRY IT 1.69 Subtract and then check by adding: $4{,}585-697$.

TRY IT 1.70 Subtract and then check by adding: $5{,}637-899$.

Translate Word Phrases to Math Notation

As with addition, word phrases can tell us to operate on two numbers using subtraction. To translate from a word phrase to math notation, we look for key words that indicate subtraction. Some of the words that indicate subtraction are listed in Table 1.3.

Operation	Word Phrase	Example	Expression
Subtraction	minus	5 minus 1	$5-1$
	difference	the difference of 9 and 4	$9-4$
	decreased by	7 decreased by 3	$7-3$
	less than	5 less than 8	$8-5$
	subtracted from	1 subtracted from 6	$6-1$

Table 1.3

EXAMPLE 1.36

Translate and then simplify:

ⓐ the difference of 13 and 8 ⓑ subtract 24 from 43

Solution

ⓐ

The word *difference* tells us to subtract the two numbers. The numbers stay in the same order as in the phrase.

	the difference of 13 and 8
Translate.	$13-8$
Simplify.	5

ⓑ

The words *subtract from* tells us to take the first number away from the second. We must be careful to get the order correct.

	subtract 24 from 43
Translate.	$43-24$
Simplify.	19

TRY IT 1.71 Translate and simplify:

ⓐ the difference of 14 and 9 ⓑ subtract 21 from 37

TRY IT 1.72 Translate and simplify:

ⓐ 11 decreased by 6 ⓑ 18 less than 67

Subtract Whole Numbers in Applications

To solve applications with subtraction, we will use the same plan that we used with addition. First, we need to determine what we are asked to find. Then we write a phrase that gives the information to find it. We translate the phrase into math notation and then simplify to get the answer. Finally, we write a sentence to answer the question, using the appropriate units.

EXAMPLE 1.37

The temperature in Chicago one morning was 73 degrees Fahrenheit. A cold front arrived and by noon the temperature was 27 degrees Fahrenheit. What was the difference between the temperature in the morning and the temperature at noon?

Solution

We are asked to find the difference between the morning temperature and the noon temperature.

Write a phrase.	the difference of 73 and 27
Translate to math notation. *Difference* tells us to subtract.	$73 - 27$
Then we do the subtraction.	$\begin{array}{r} {}^{6}\not{7}\,{}^{13}\not{3} \\ -\ 2\,7 \\ \hline 4\,6 \end{array}$
Write a sentence to answer the question.	The difference in temperatures was 46 degrees Fahrenheit.

TRY IT 1.73 The high temperature on June 1st in Boston was 77 degrees Fahrenheit, and the low temperature was 58 degrees Fahrenheit. What was the difference between the high and low temperatures?

TRY IT 1.74 The weather forecast for June 2 in St Louis predicts a high temperature of 90 degrees Fahrenheit and a low of 73 degrees Fahrenheit. What is the difference between the predicted high and low temperatures?

EXAMPLE 1.38

A washing machine is on sale for $399. Its regular price is $588. What is the difference between the regular price and the sale price?

Solution

We are asked to find the difference between the regular price and the sale price.

Write a phrase.	the difference between 588 and 399
Translate to math notation.	$588 - 399$
Subtract.	$\begin{array}{r} {}^{4}\not{5}\,{}^{17}\not{8}\,{}^{18}\not{8} \\ -\ 3\,9\,9 \\ \hline 1\,8\,9 \end{array}$
Write a sentence to answer the question.	The difference between the regular price and the sale price is $189.

TRY IT 1.75 A television set is on sale for $499. Its regular price is $648. What is the difference between the regular price and the sale price?

TRY IT 1.76 A patio set is on sale for $149. Its regular price is $285. What is the difference between the regular price and the sale price?

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Model subtraction of two-digit whole numbers (https://www.openstax.org/l/24sub2dignum)
Model subtraction of three-digit whole numbers (https://www.openstax.org/l/24sub3dignum)
Subtract Whole Numbers (https://www.openstax.org/l/24subwholenum)

SECTION 1.3 EXERCISES

Practice Makes Perfect

Use Subtraction Notation

In the following exercises, translate from math notation to words.

141. $15 - 9$ **142.** $18 - 16$ **143.** $42 - 35$

144. $83 - 64$ **145.** $675 - 350$ **146.** $790 - 525$

Model Subtraction of Whole Numbers

In the following exercises, model the subtraction.

147. $5 - 2$ **148.** $8 - 4$ **149.** $6 - 3$

150. $7 - 5$ **151.** $18 - 5$ **152.** $19 - 8$

153. $17 - 8$ **154.** $17 - 9$ **155.** $35 - 13$

156. $32 - 11$ **157.** $61 - 47$ **158.** $55 - 36$

Subtract Whole Numbers

In the following exercises, subtract and then check by adding.

159. $9 - 4$ **160.** $9 - 3$ **161.** $8 - 0$

162. $2 - 0$ **163.** $38 - 16$ **164.** $45 - 21$

165. $85 - 52$ **166.** $99 - 47$ **167.** $493 - 370$

168. $268 - 106$ **169.** $5{,}946 - 4{,}625$ **170.** $7{,}775 - 3{,}251$

171. $75 - 47$ **172.** $63 - 59$ **173.** $461 - 239$

174. $486 - 257$ **175.** $525 - 179$ **176.** $542 - 288$

177. $6{,}318 - 2{,}799$ **178.** $8{,}153 - 3{,}978$ **179.** $2{,}150 - 964$

180. $4{,}245 - 899$ **181.** $43{,}650 - 8{,}982$ **182.** $35{,}162 - 7{,}885$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate and simplify.

183. The difference of 10 and 3 **184.** The difference of 12 and 8 **185.** The difference of 15 and 4

186. The difference of 18 and 7 **187.** Subtract 6 from 9 **188.** Subtract 8 from 9

189. Subtract 28 from 75 **190.** Subtract 59 from 81 **191.** 45 decreased by 20

192. 37 decreased by 24 **193.** 92 decreased by 67 **194.** 75 decreased by 49

195. 12 less than 16 **196.** 15 less than 19 **197.** 38 less than 61

198. 47 less than 62

Mixed Practice

In the following exercises, simplify.

199. $76 - 47$

200. $91 - 53$

201. $256 - 184$

202. $305 - 262$

203. $719 + 341$

204. $647 + 528$

205. $2{,}015 - 1{,}993$

206. $2{,}020 - 1{,}984$

In the following exercises, translate and simplify.

207. Seventy-five more than thirty-five

208. Sixty more than ninety-three

209. 13 less than 41

210. 28 less than 36

211. The difference of 100 and 76

212. The difference of 1,000 and 945

Subtract Whole Numbers in Applications

In the following exercises, solve.

213. **Temperature** The high temperature on June 2 in Las Vegas was 80 degrees and the low temperature was 63 degrees. What was the difference between the high and low temperatures?

214. **Temperature** The high temperature on June 1 in Phoenix was 97 degrees and the low was 73 degrees. What was the difference between the high and low temperatures?

215. **Class size** Olivia's third grade class has 35 children. Last year, her second grade class had 22 children. What is the difference between the number of children in Olivia's third grade class and her second grade class?

216. **Class size** There are 82 students in the school band and 46 in the school orchestra. What is the difference between the number of students in the band and the orchestra?

217. **Shopping** A mountain bike is on sale for $399. Its regular price is $650. What is the difference between the regular price and the sale price?

218. **Shopping** A mattress set is on sale for $755. Its regular price is $1,600. What is the difference between the regular price and the sale price?

219. **Savings** John wants to buy a laptop that costs $840. He has $685 in his savings account. How much more does he need to save in order to buy the laptop?

220. **Banking** Mason had $1,125 in his checking account. He spent $892. How much money does he have left?

Everyday Math

221. **Road trip** Noah was driving from Philadelphia to Cincinnati, a distance of 502 miles. He drove 115 miles, stopped for gas, and then drove another 230 miles before lunch. How many more miles did he have to travel?

222. **Test Scores** Sara needs 350 points to pass her course. She scored 75, 50, 70, and 80 on her first four tests. How many more points does Sara need to pass the course?